Lecture Scribe: Tails & Inequalities

L21_L22 (1).pdf
PDF

Generate a lecture scribe intended to serve as faithful, exam-ready reference material for revision.

Use only the provided context consisting of the lecture slides and relevant textbook excerpts. Do not introduce any material that does not appear in the given context.

While writing the scribe, reason step by step through all definitions, notation, assumptions, theorems or results, proofs or derivations, and worked example solutions exactly as presented in the context.

Make the logical flow between concepts explicit so that a student can clearly follow the reasoning during exam preparation.

Maintain a formal academic tone and structure the scribe clearly for revision purposes.

Below is a **faithful, exam-ready lecture scribe** based strictly on your uploaded slides . I have preserved definitions, notation, intuition, examples, and logical flow exactly as presented, while making the reasoning explicit for revision.

Lecture 21–22: Inequalities and Tail Bounds

(CSE400 – Fundamentals of Probability in Computing)

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1. Introduction and Motivation

1.1 What is a Tail?

Definition:
The *tail* of a random variable X is:

$$P(X > x)$$

This represents the probability that the random variable takes **large values beyond a threshold** x.

1.2 Why Do We Care About Tails?

From the slides (page 5):

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

These represent **rare but critical events**.

1.3 Key Idea

- **Mean → average behavior (easy to compute)**
- **Tail → rare extreme events (hard to compute)**

Thus, while averages describe normal behavior, **tails capture worst-case scenarios**, which are crucial in systems.

1.4 Why Are Tails Hard to Compute?

- Requires **summing many small probabilities**
- Leads to **long and complex expressions**

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2. Examples of Tail Events

2.1 Example 1: Load Balancing (Binomial Setting)

Problem:
Distribute n jobs among n servers randomly.

Find:

$$P(\text{a server gets } \geq k \text{ jobs})$$

Intuition (from slides):

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a **tail event (rare overload)**

Mathematical Model:

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

2.2 Example 2: Poisson Arrivals

Problem:
Jobs arrive at rate λ (Poisson process).
Find:

$$P(\text{at least } k \text{ arrivals in 1 hour})$$

Intuition:

- Jobs arrive randomly over time
- Most hours → moderate arrivals
- Sometimes → burst of arrivals

This is a **tail event (sudden spike)**

Mathematical Model:

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

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3. Why Tail Bounds Are Needed

3.1 Exact Computation is Difficult

From slides (page 14):

- Binomial and Poisson tails require **long summations**
- These are:
  - Computationally expensive
  - Hard to simplify

3.2 Key Idea: Approximate Instead of Exact

Instead of computing:

$$P(X \geq k)$$

We compute an **upper bound**:

$$P(X \geq k) \leq \text{simple expression}$$

3.3 Advantages

- Easier to compute
- Still provides a **guarantee**
- Useful in system design

3.4 Core Philosophy

> **"Approximate safely instead of computing exactly."**

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4. Running Example (Used Across Inequalities)

We analyze all bounds using the same setup:

Experiment:

- Flip a fair coin n times

Random Variable:

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean:

$$E[X] = \frac{n}{2}$$

Tail Question:

$$P\left(X \geq \frac{3n}{4}\right)$$

This represents **getting unusually many heads**, i.e., a **tail region**.

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5. Markov's Inequality

5.1 Key Idea

- Uses **only the mean**
- Large values are **rare**

Example intuition:

- If average = 10 → very few values can be ≥ 100

5.2 Statement of Markov's Inequality

For any non-negative random variable X and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

5.3 Interpretation

- Probability of large deviation is bounded using **mean only**
- No need for full distribution

5.4 Logical Flow

1. We want $P(X \geq a)$
2. Exact computation is hard
3. Use only $E[X]$
4. Obtain a simple upper bound

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6. Overall Conceptual Flow (Important for Exams)

1. **Tail definition:**
   $P(X > x)$
2. **Motivation:**
   Rare events (failures, overloads)
3. **Problem:**
   Exact computation requires large sums
4. **Solution:**
   Use **tail bounds**
5. **First tool:**
   Markov's Inequality
6. **Key principle:**
   Use simple statistics (like mean) to bound complex probabilities

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7. Key Takeaways

- Tail probabilities measure **extreme events**
- Exact computation is **difficult due to summations**
- Tail bounds provide:
  - Simplicity
  - Guarantees
- Markov inequality is the **simplest bound**
- More advanced bounds (Chebyshev, Chernoff) improve tightness (as per outline)

If you want, I can continue this scribe with:

- **Full proof of Markov inequality**
- **Chebyshev + Chernoff derivations**
- **Step-by-step solved examples (exam-style)**